

Autonom Robot med Python og AI

Made by Zuhair Haroon Khan

Robots in Science Fiction



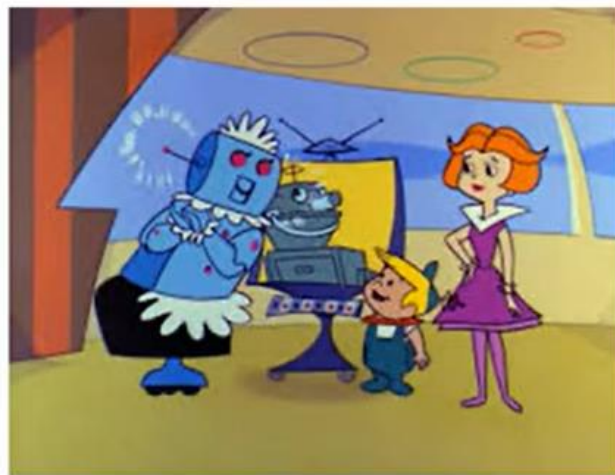
Q the Automaton, 1918



Metropolis, 1927



Star Wars, 1977



The Jetsons, 1962



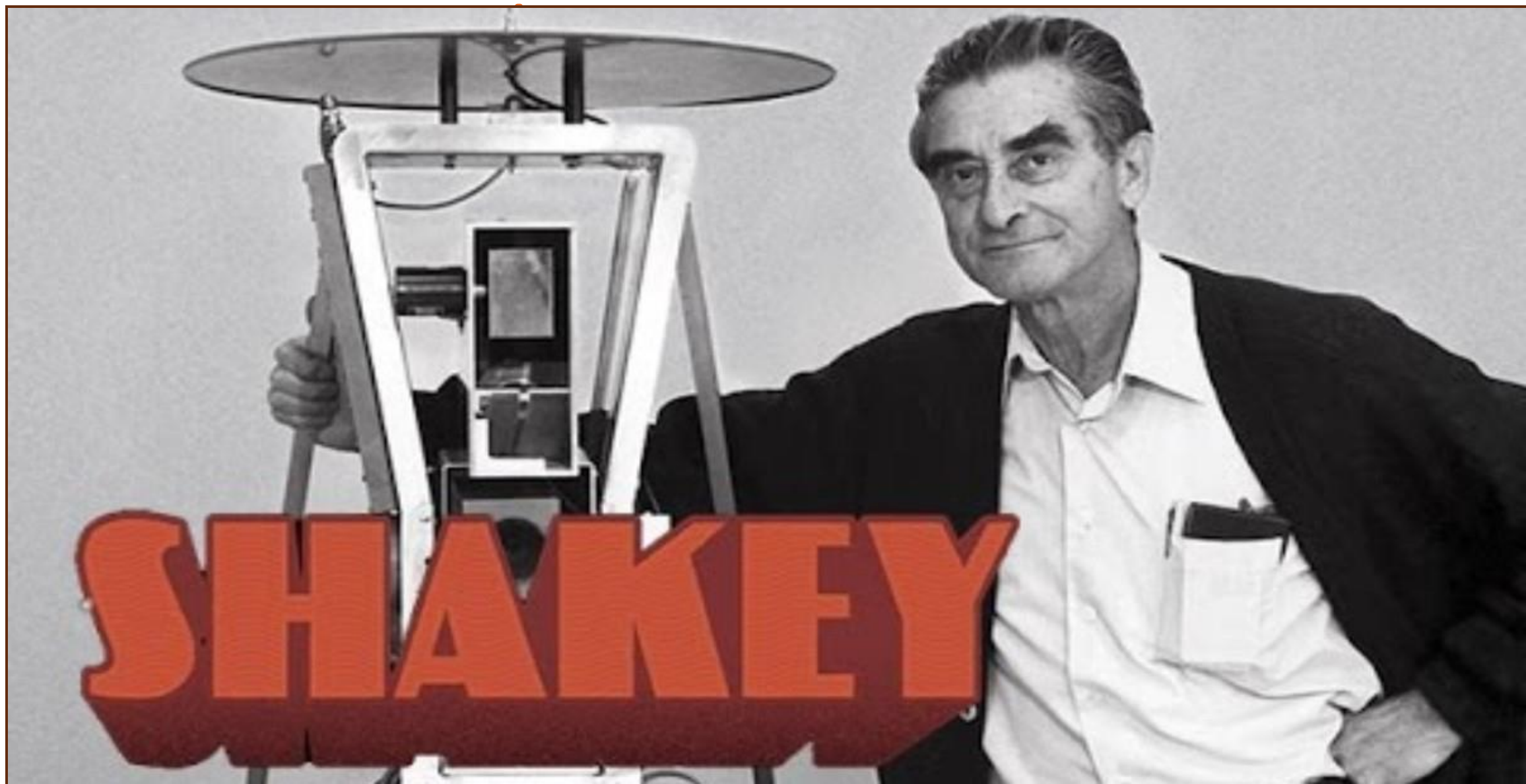
Transformers, 2007



Wall-E, 2008

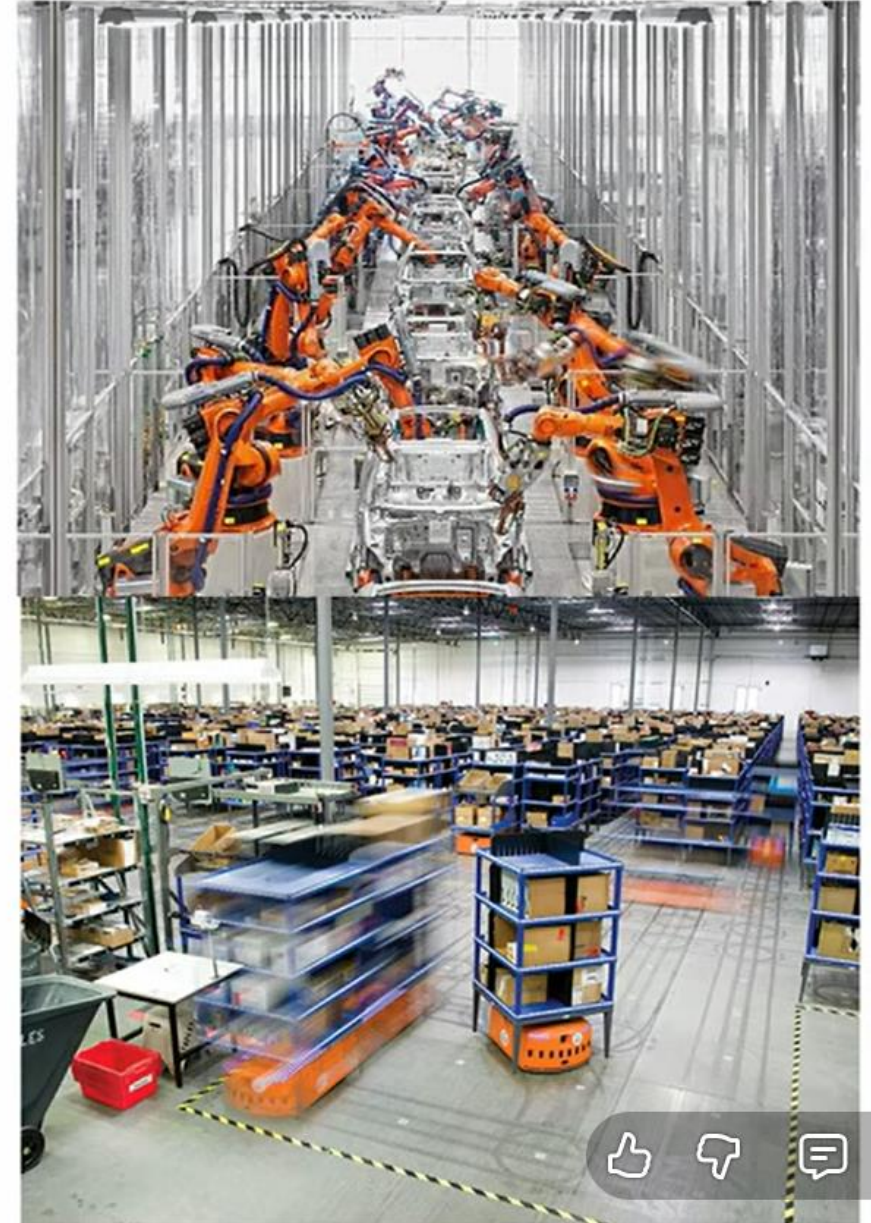
Why Robots





Robot Engineering

- Sense-Plan-Act
- Similar to Shakey (1966)
- Structured Environments



When State Estimation Fails



DARPA Robotics Challenge 2015



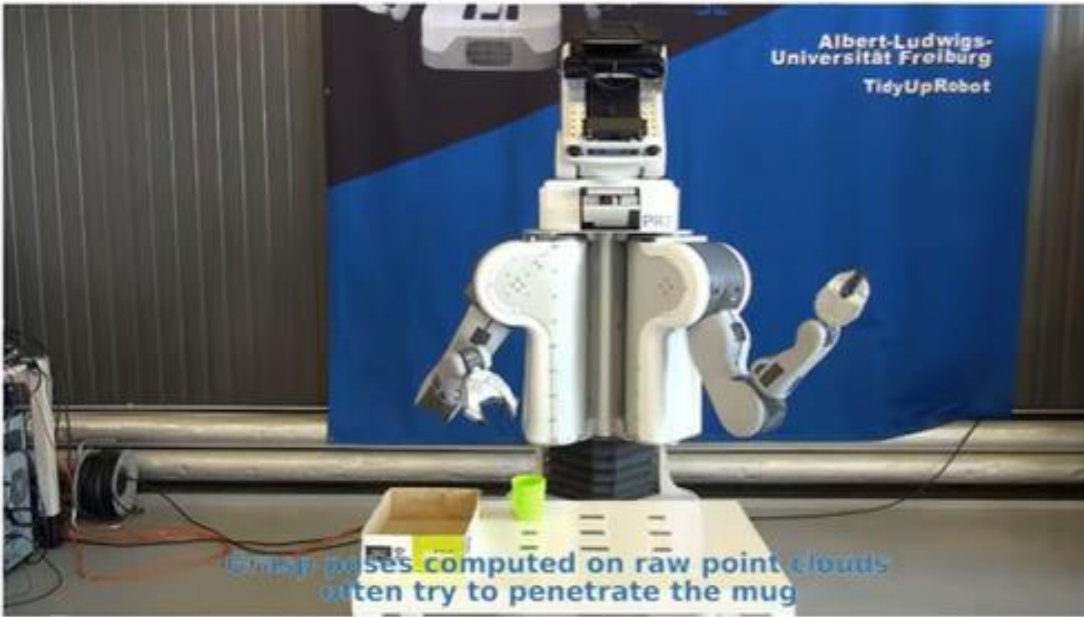
Boston Dynamics 2018



Boston Dynamics 2022

Moravec's Paradox (1988)

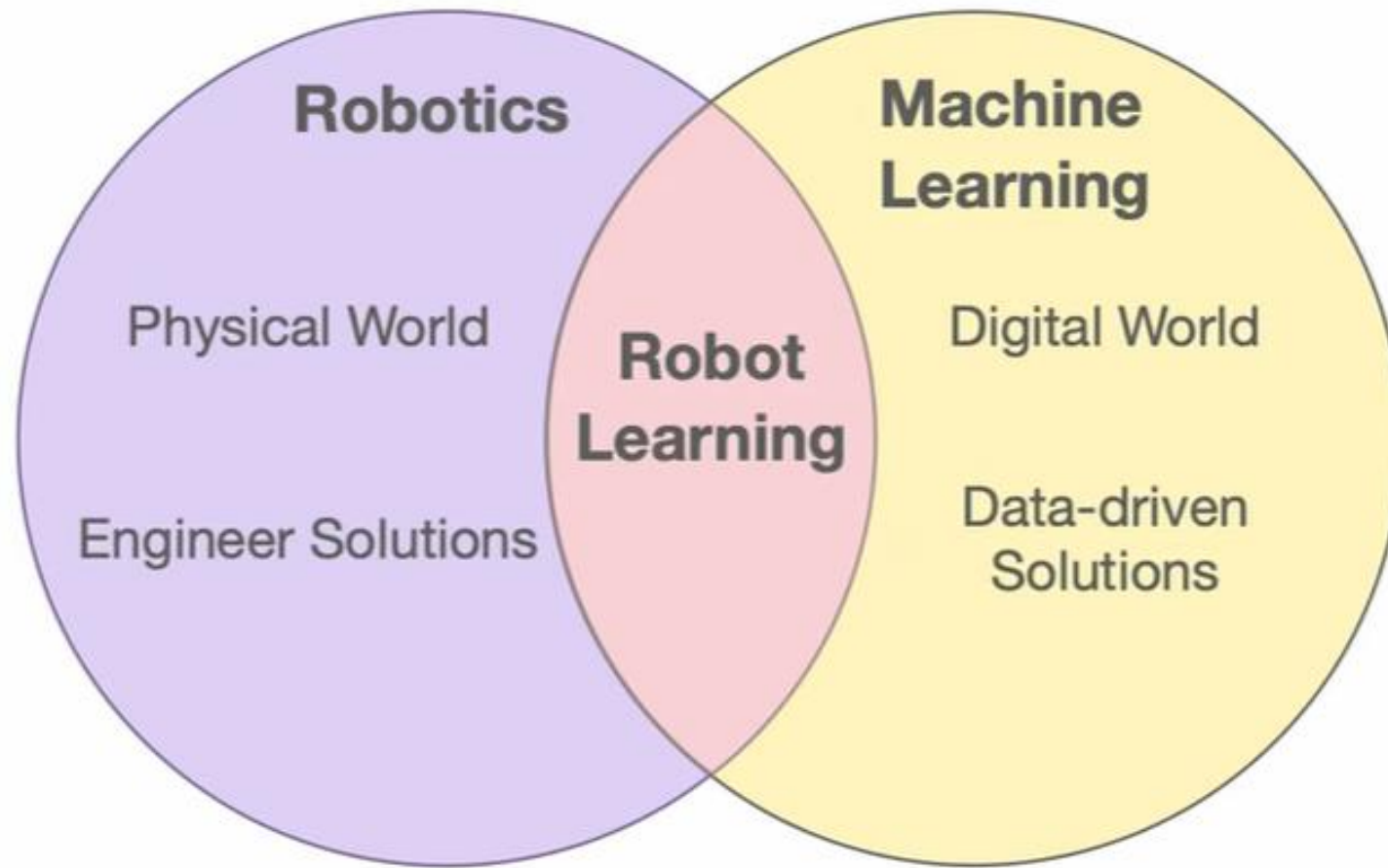
- **Abstract thinking:** hard for animals, easy for AI
- **Sensorimotor skills:** easy for animals, hard for AI



Mees et al., 2019



Robot Learning



Robot Learning

Solving **Robotics** via **Machine Learning**



Perception
Control
...



Imitation Learning
Reinforcement Learning
Dynamics Learning
Representation Learning
...

Are These Robots?



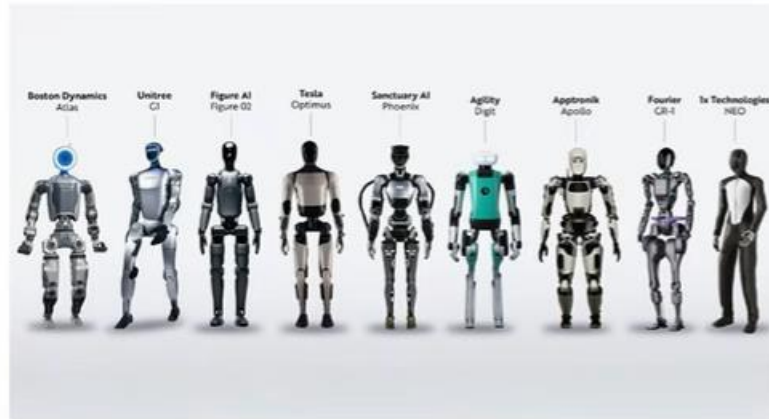
Why not we
have
autonomous
robot

Why Don't We Already Have **Autonomous Robots?**



Berger and Wyrobeck, Stanford 2007

Recent Hardware Advances



Humanoid Robots



NVIDIA GPUs

Recent AI Advances: **Data Scaling**

- Avg time for a human to read dataset

Imagenet



2 years

GPT-2



60 years

Llama 3

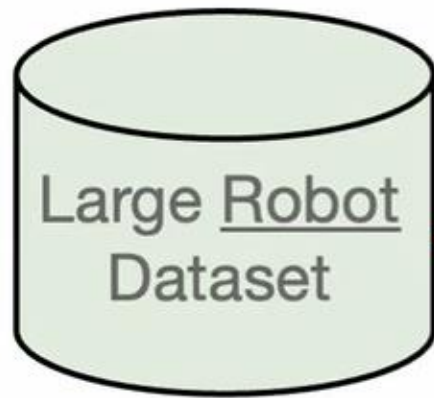
90,000 years

Recent AI Advances: **Models**

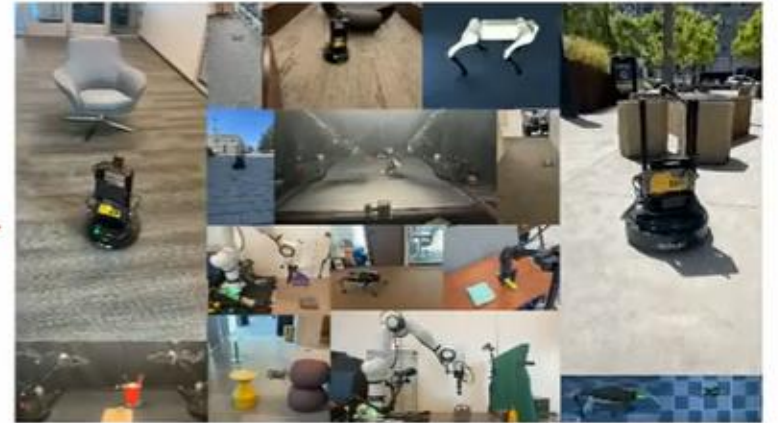


Goal: Generalist Robot Policies

Where do we get this data from?

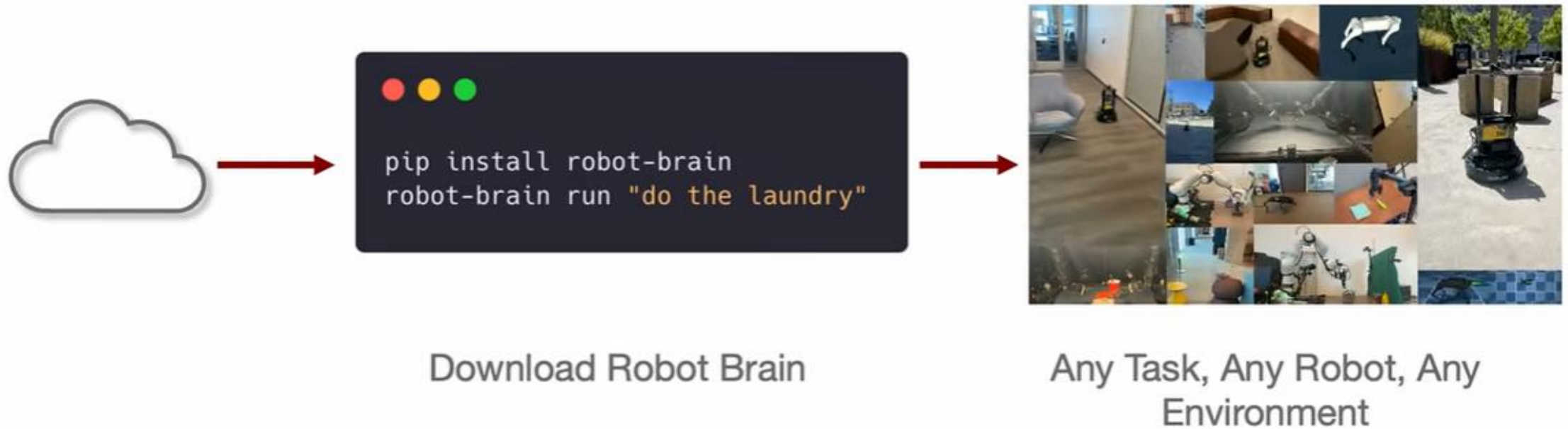


How can we model the data distribution?

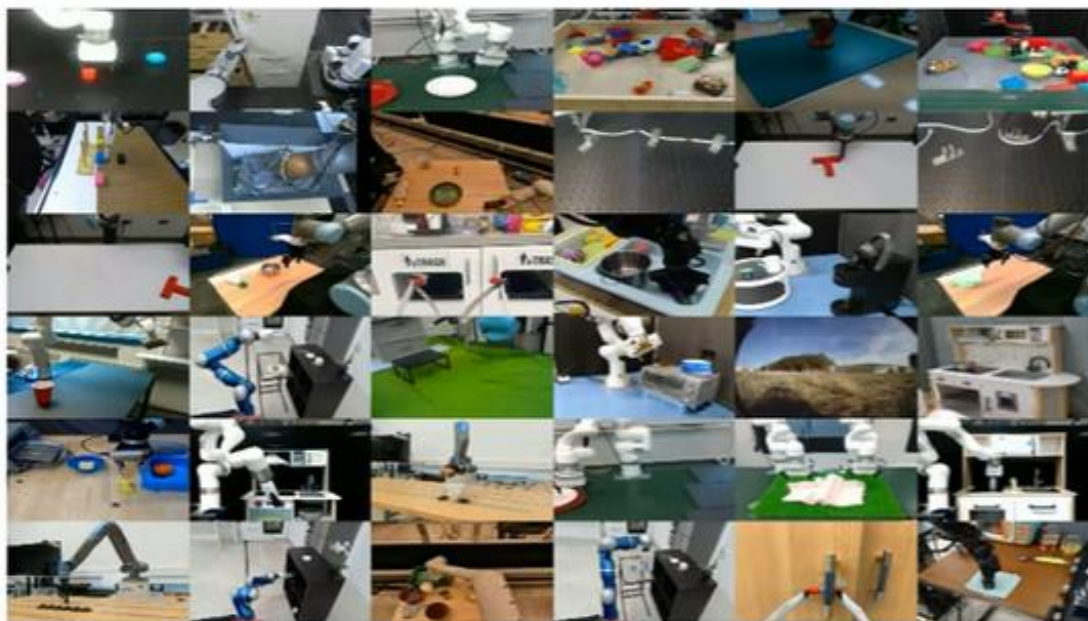


Any Task, Any Robot, Any Environment

Goal: **Plug-and-Play** Generalist Robot Policies



Large Datasets



Open X-Embodiment: Robotic Learning Datasets and RT-X Models

Open X-Embodiment Collaboration, Mees et al., ICRA, 2024.

Best Conference Paper Award (out of 1765 papers)

Large Models



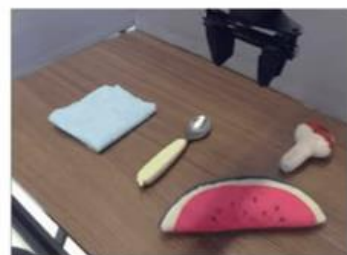
Octo

Isaac GROOT

π_0

RT-2

...



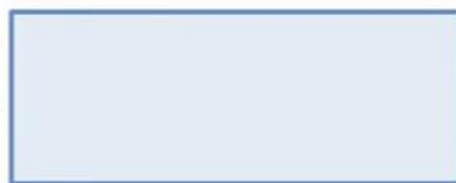
"Pick up the
spoon"

Policy



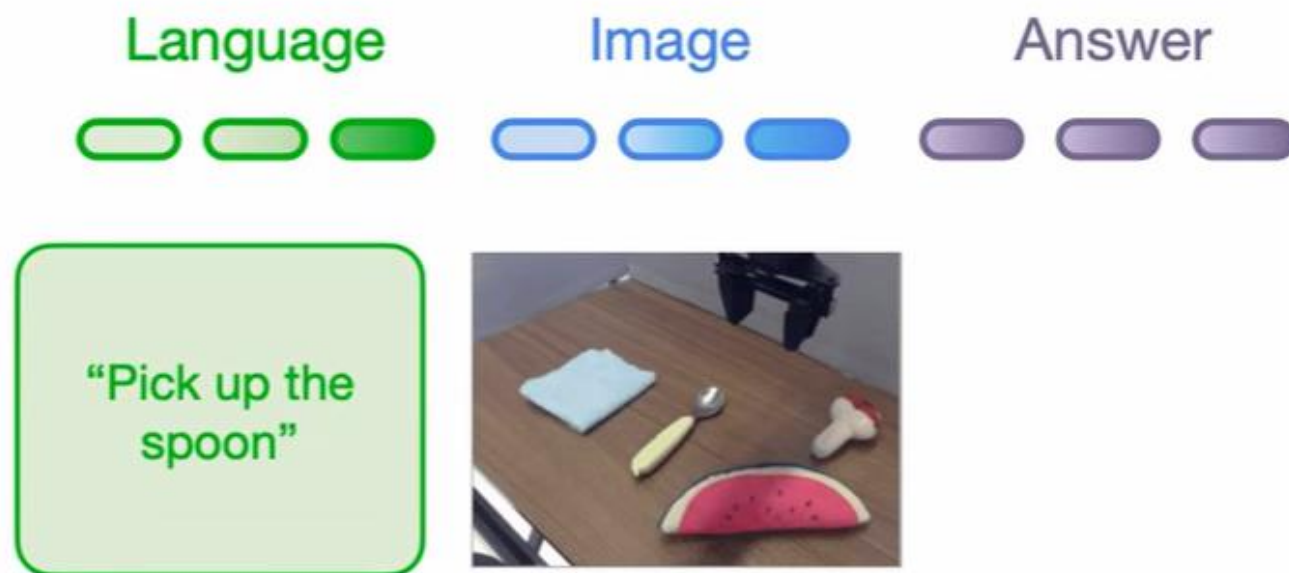


“Caption the scene”

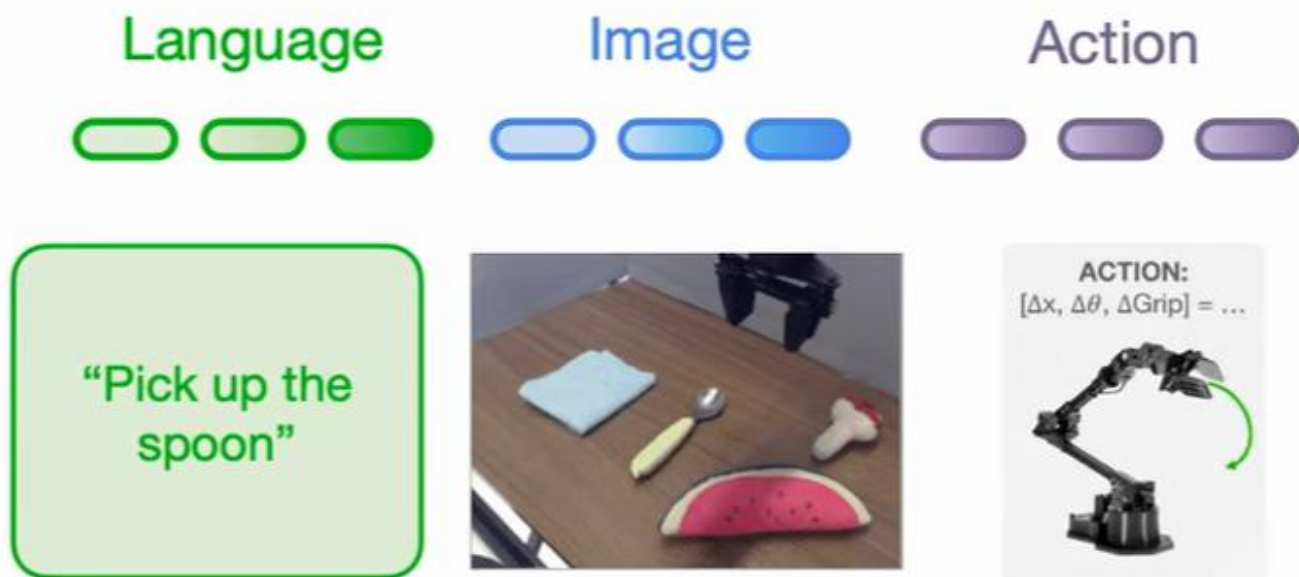


“The picture shows the Statue of Liberty in NY”

Key: Robotics as Multimodal Sequence Modeling



Key: Robotics as Multimodal Sequence Modeling



**So we train a large transformer on robot data
and we are done?**

Large Datasets

Robot Data is **Scarce**

Collecting Data
requires **Human
Supervision**

Large Models

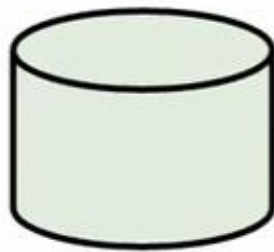
Robot Data is
**Multimodal &
Heterogeneous**

Robots need **High
Frequency Control**

Scalable Evaluation

Robot Evals are
Tedious & Expensive

Difficult to reproduce



Algorithms for Robotic Learning

Imitation Learning

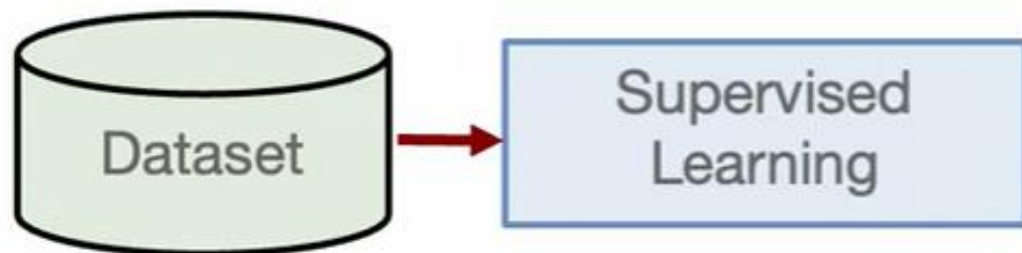
Given labeled data:

$$\mathcal{D} = \{(x_i, y_i)\},$$

learn $f(x) \approx y$

Assumes:

inputs x are independently,
identically distributed (i.i.d.)

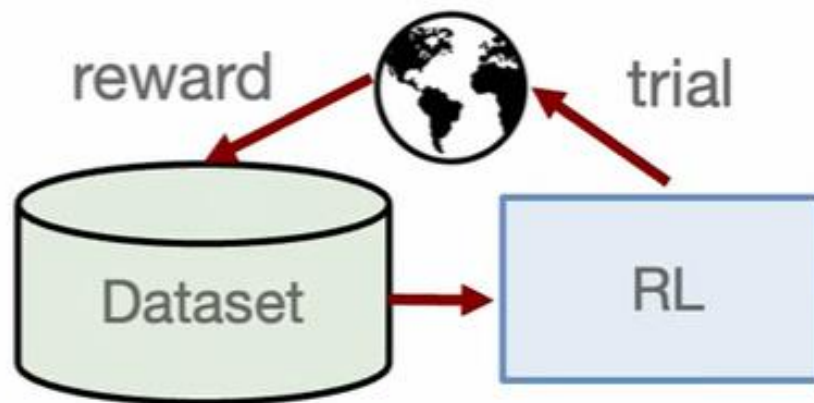


Reinforcement Learning

Learn behavior:

$$\pi(a|s)$$

data is **not** i.i.d., actions
affect future states



Why You Should Study Robot Learning

- Broad skills from foundation model training to PID control
- Advances often applicable to all robotic applications



References

- <https://www.youtube.com/watch?v=90raNpc11tQ&t=22s>